



Adapted from
*Understanding
Science*,
UC Berkeley,
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Figure 1-5

Exploration and Discovery

Ideas in science can arise in many ways—from simple curiosity or from the need to solve a particular problem. Scientists often begin investigating by making observations, asking questions, talking with colleagues, and reading about previous experiments.

READING TOOL

Copy one of the concept maps in this lesson. Then, add specific examples to illustrate the processes it describes.

Exploration and Discovery

Usually, the testing of ideas begins with observations and questions. Those observations and questions come from a variety of different sources, as shown in **Figure 1-5**.

Scientific Attitudes Good scientists share scientific attitudes, or habits of mind, that lead them to exploration and discovery.

Curiosity, skepticism, open-mindedness, and creativity help scientists and engineers ask new questions and define new problems.

Curiosity A scientist may look at a salt marsh and ask, “What’s that plant? Why is it growing here?” Previous studies can spark curiosity and lead to new questions. Engineering problems often arise from practical issues involving humans and health or environmental issues.

Skepticism Scientists and engineers should be skeptics who question existing ideas and hypotheses, and refuse to accept explanations without evidence. Scientists often design new experiments to test hypotheses proposed by other scientists. Engineers work to solve problems in the physical world using scientific thinking.

Open-Mindedness Scientists and engineers should be open-minded, meaning that they are willing to accept different ideas that may not agree with their hypotheses.

Creativity Creativity and critical thinking are essential for asking questions, defining problems, proposing hypotheses, and designing experiments that yield accurate data.